



Repowering the Plains Empowering the People

Our Team

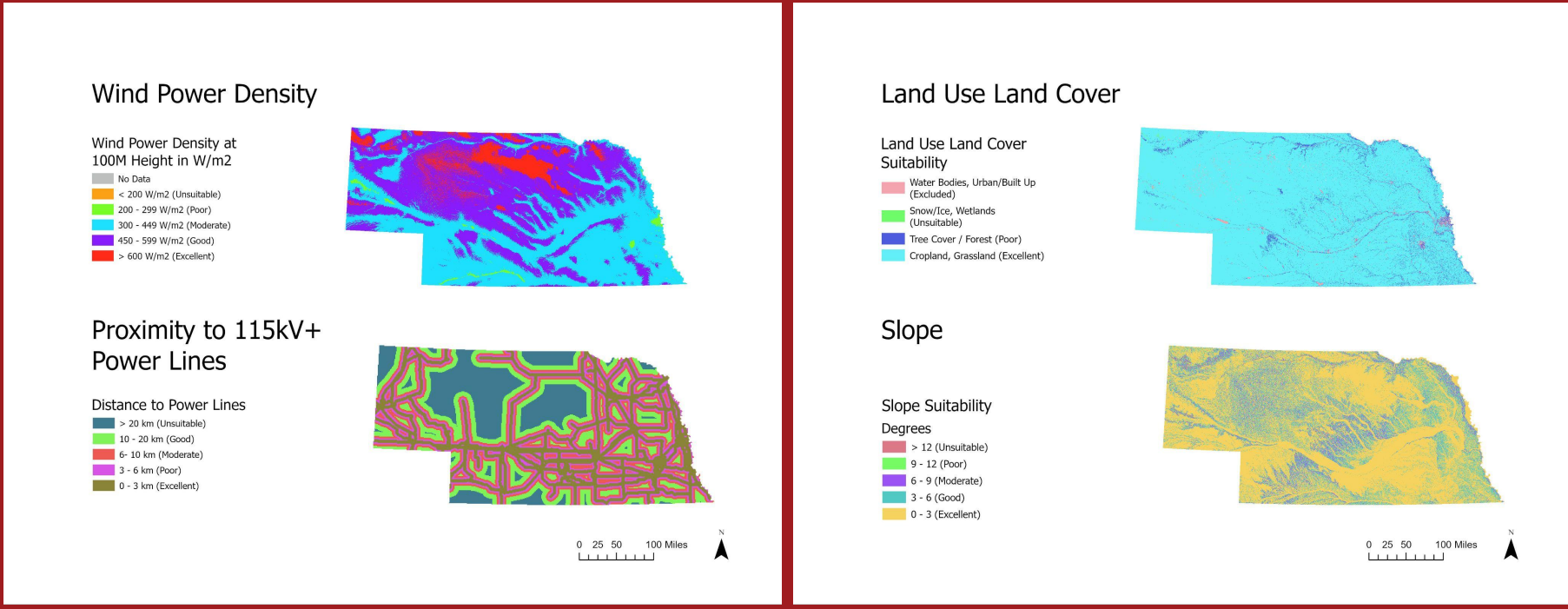


Trojan Wind is USC's inaugural Collegiate Wind Competition team - 9 students across 7 majors and 3 schools. We collectively believe in empowering community-centric energy that is renewable, socially beneficial, and efficient. Wind energy perfectly positions itself to fulfill these requirements, and we believe it plays an integral role in achieving a green future.

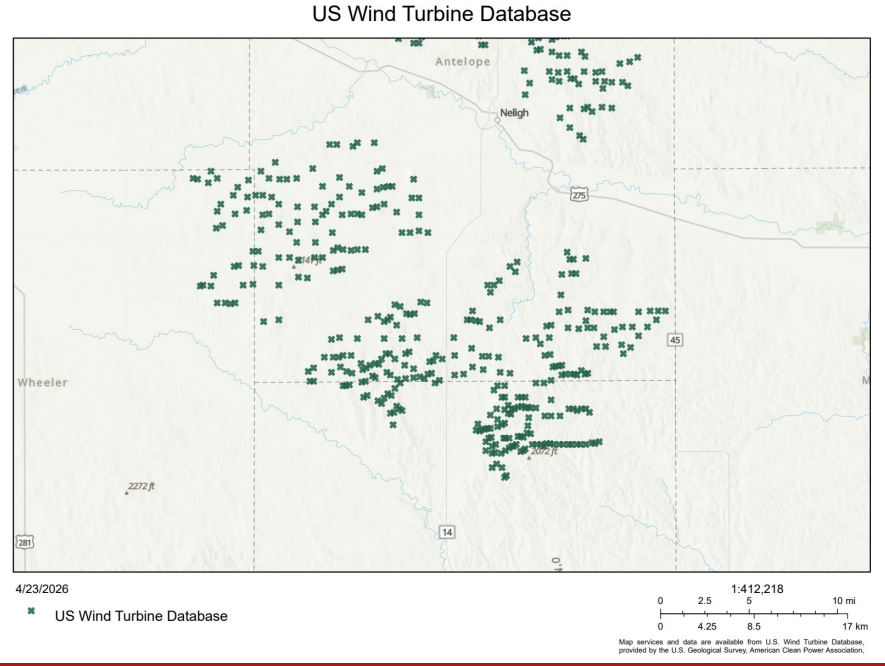
Motivation

Nebraska holds the highest utility-scale wind resource density in the U.S. yet installed capacity represents only a fraction of its technical potential. We aimed to contribute to the realization of it potential by operating as an independent project developer. We applied a rigorous five-phase geospatial methodology to site a 72 MW wind farm in Antelope County, Nebraska: 10 × Vestas V162-7.2 MW turbines, ~325 GWh/yr net AEP, and a leveraged LCOE of \$43.88/MWh - within NREL's Great Plains benchmark range.

Site Selection Method



Variable	Importance Score
Wind Power Density	0.40
Distance to Transmission Lines (≥115 kV)	0.31
Slope	0.10
Distance to Major Roads	0.09
Bare/Sparse Vegetation (<15% cover)	0.08
Grassland: Herbaceous Vegetation	0.01



Trojan Wind developed a five-phase geospatial methodology to narrow Nebraska's 50 million acres to a single site in Antelope County.

Phase 1: Multi-Criteria Decision Analysis

We collected five spatial variables statewide in ArcGIS Pro: wind power density, proximity to 115kV+ transmission, land use/land cover, slope, and road distance.

Phase 2: ML-Derived Weighting

We replaced analyst judgment with data-driven importance scores by deriving variable weights from a Random Forest model trained on 1,234 existing U.S. wind farm locations. Wind power density and transmission proximity were found to be most important, cumulatively accounting for 71% of predictive weight.

Phase 3: Exclusionary Masking

We applied buffers around FAA, military, wetland, and protected-area. Thus reducing the 50 million acres to 23 candidate sites across north-central Nebraska.

Phase 4: Policy Screening

We screened the 23 sites across 17 counties for ordinances, social acceptance, military overlaps, and protected areas. 5 counties (Brown, Holt, Antelope, Wayne, and Custer) survived with no disqualifying flags.

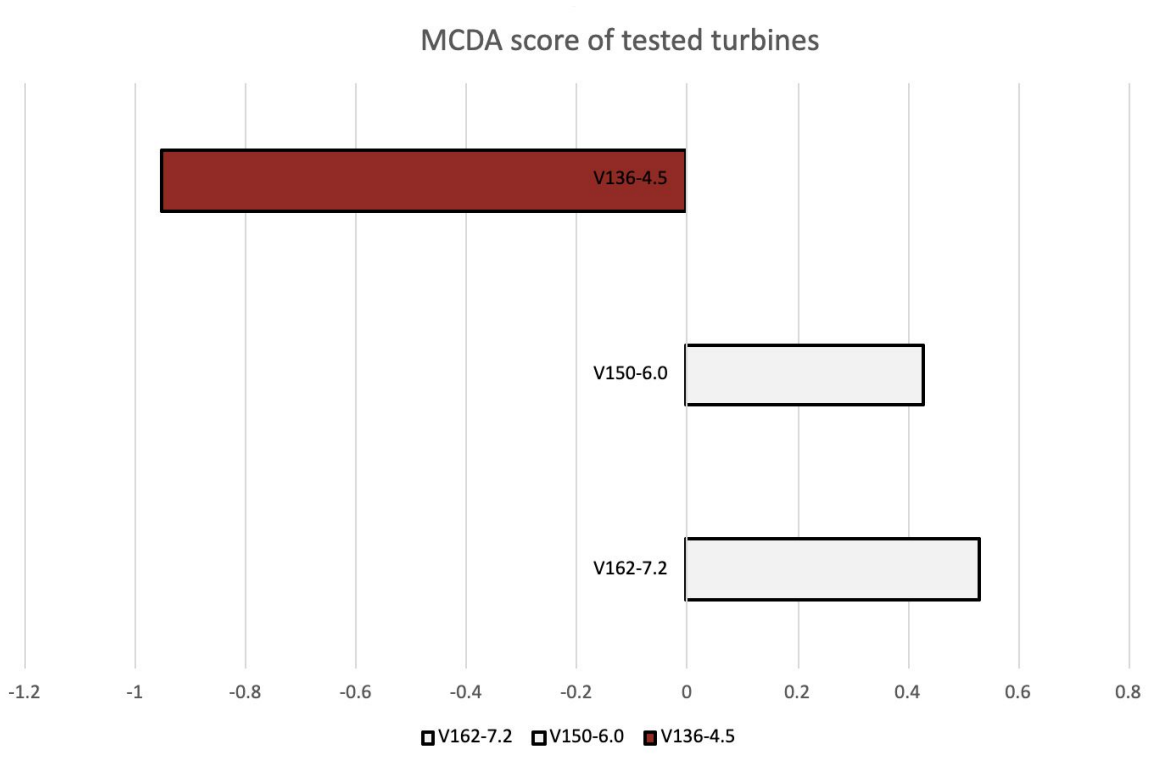
Phase 5: Final Selection

Antelope County was selected based on its: top-quartile suitability, 450–600 W/m² wind power density, two 150 kV+ transmission lines on-site, no active ordinance restrictions, and sufficient separation from the whooping crane Central Flyway corridor.

Turbine Selection

platform	n_turbines	cf_pct	aep_net_gwh	total_capex_musd	lcoe_usd_mwh	npv_musd	mnda_score
V162-7.2	18	58.05	315.68	122.96	44.79	-45.82	0.53
V158-6.0	12	49.82	389.19	128.73	45.16	-45.37	0.43
V136-4.5	16	45.95	289.81	118.53	47.67	-50.62	-0.95

WINNER: V162-7.2 × 18
LCOE: \$44.79/MWh | NPV (unsub, PPA=\$32): \$-45.0M | CF: 58.1%

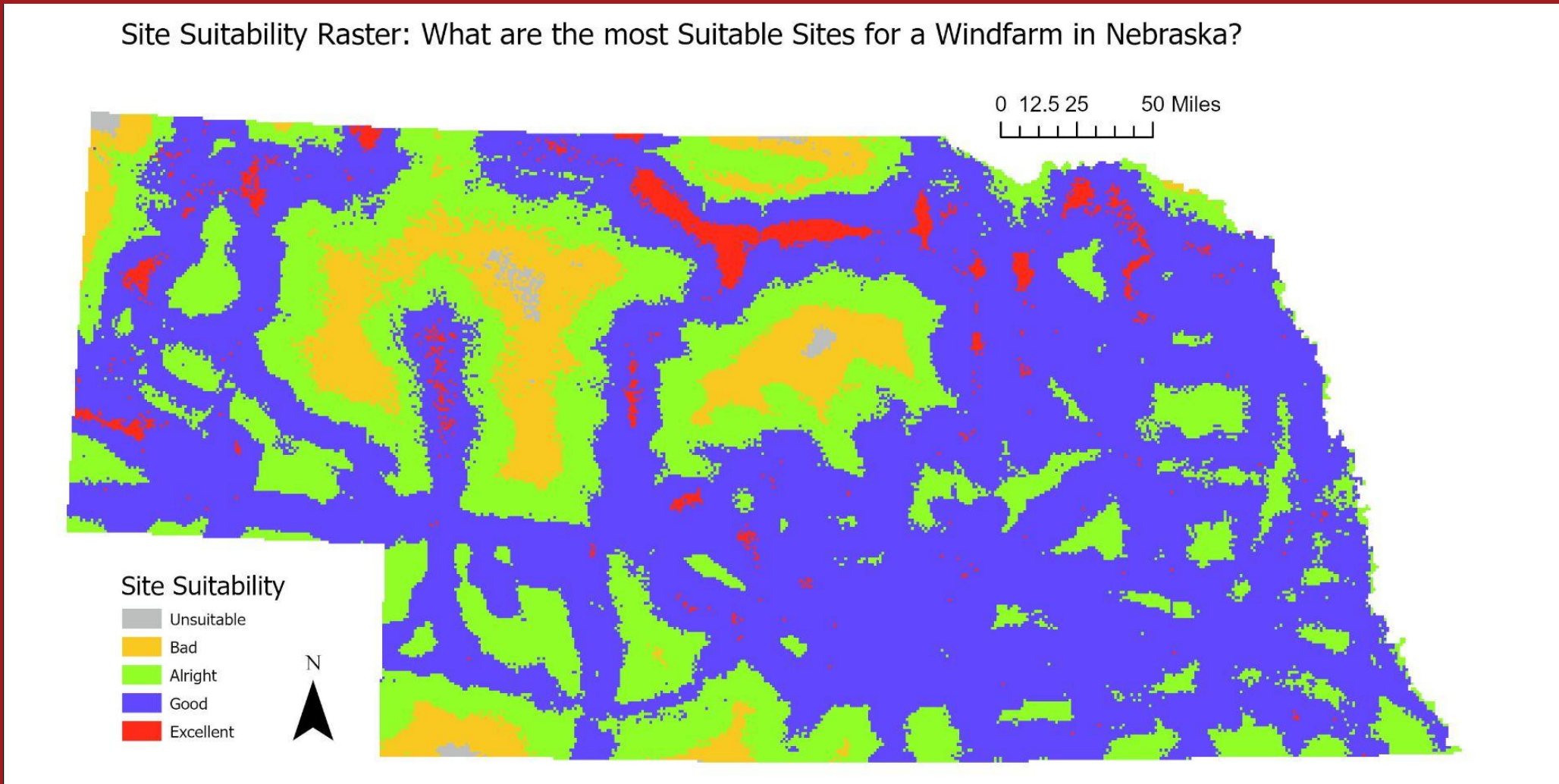


The Vestas V162-7.2 MW was selected via a 6-criterion MCDA evaluated against comparable Class II and Class S platforms. Key advantages:

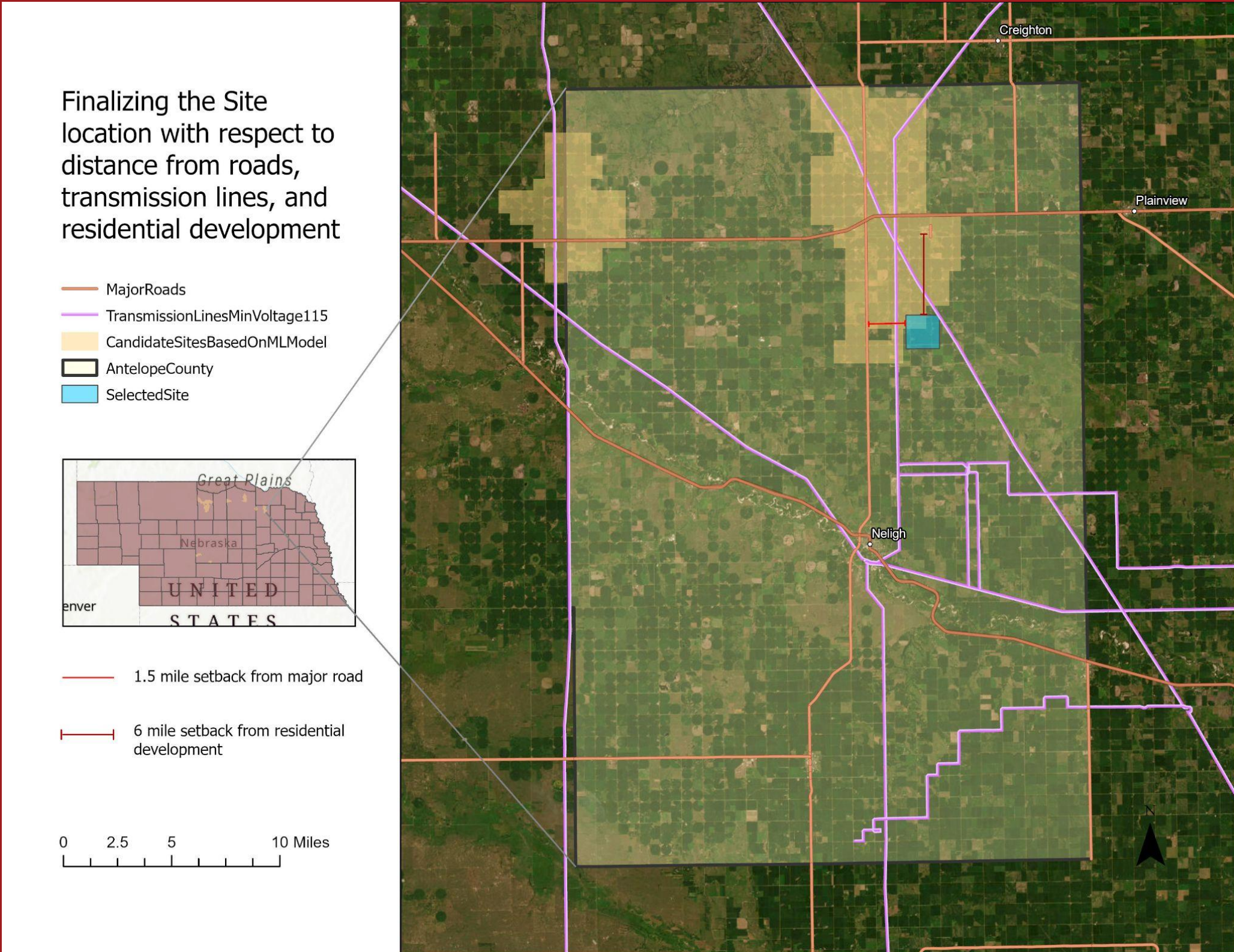
- 162 m rotor diameter and 119m hub height maximise AEP at Nebraska's 7–8 m/s wind regime
- Site-specific customisation
 - cold-climate operation to -30°C with integrated de-icing system
 - bat deterrent system reduces mortality by up to 78%
 - Vestas' Colorado manufacturing minimises tariff exposure.

10 turbines were selected, representing the maximum integer count within the Antelope County CUP framework while remaining below the 80 MW PURPA threshold.

Site Selection Results

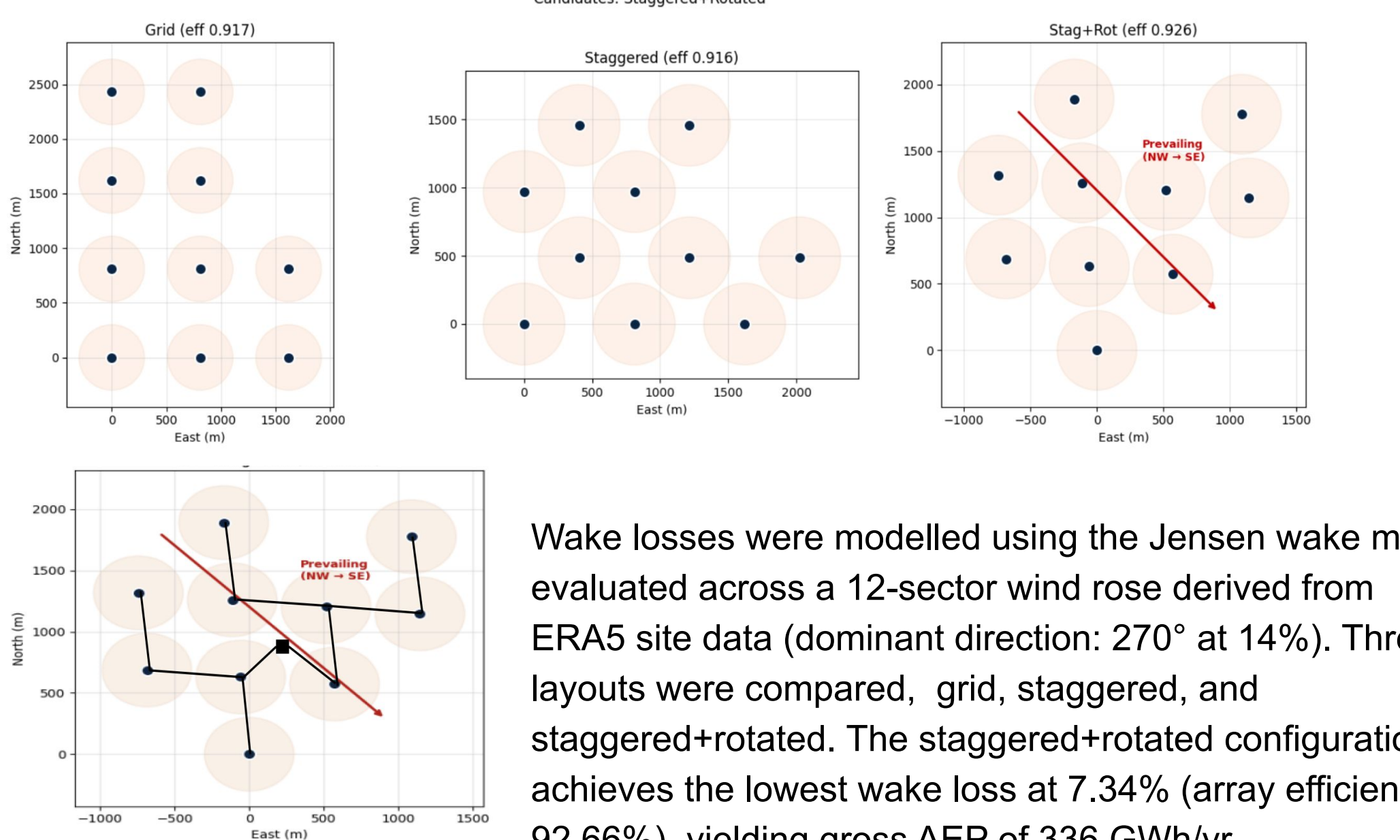


The ML-weighted suitability raster reveals that Nebraska's highest-scoring sites concentrate in two distinct regions: the north-central Sandhills corridor, where wind power density peaks above 500 W/m², and a secondary band across the southwestern Panhandle. Antelope County sits within the Sandhills high-suitability band, scoring in the top quartile statewide across all five weighted variables.



The selected site is entirely on active farmland, preserving agricultural use through co-location - turbine lease payments supplement rather than displace farm income. The nearest township is more than 6 km away, keeping noise levels well below Nebraska's residential limit and minimizing visual impact on residents. A 150 kV transmission line runs directly through the site boundary, eliminating the need for costly spur construction.

Layout Optimization



Three final AEP adjustments were applied: ~2.2% cold-climate ice loss, partially offset by the Vestas de-icing system recovering 70% of losses; +3.0% digital twin uplift; and AI SmartCurtailment reducing biological curtailment from 3.5% to 1.5%.

Net AEP: ~325 GWh/yr at 51.6% capacity factor.

Financial Model

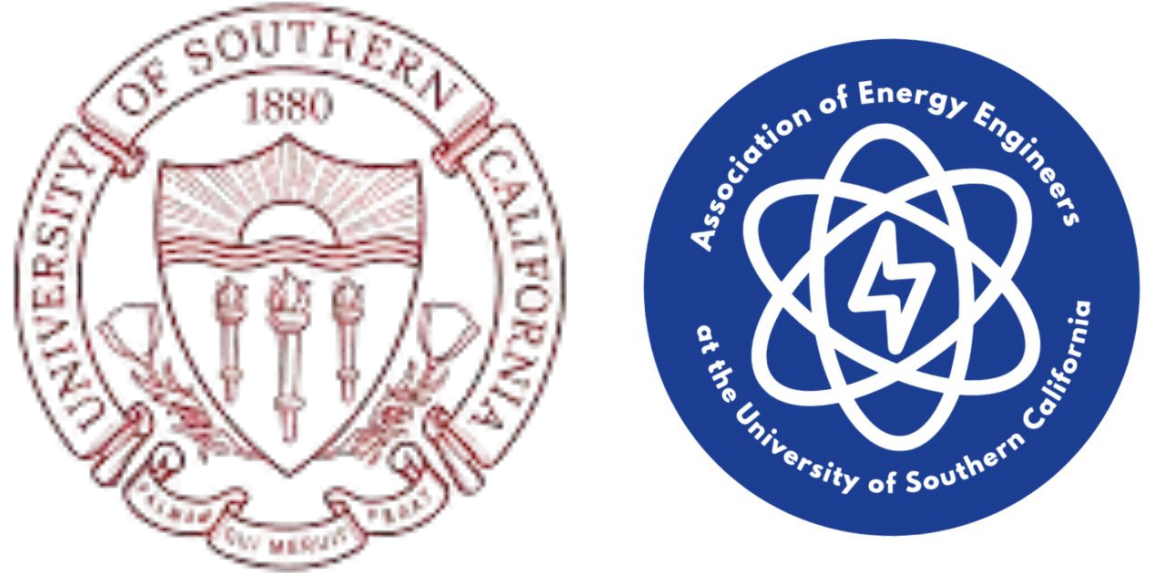
25-Year Cash Flow Schedule							
Trojan Wind, 72 MW, Antelope County NE. All figures in \$M nominal except AEP (GWh).							
	PTC active (Yr 1-10)	DEP retired (Yr 5)	PTC cliff (Yr 11)				
Yr	AEP (GWh)	Revenue	OpEx	EBITDA	MACRS Shield	CFADS	Debt Svc
1	296.8	\$29.5	\$5.2	\$24.3	\$3.8	\$28.1	\$20.8
2	295.3	\$29.5	\$5.3	\$24.2	\$7.0	\$31.1	\$23.1
3	293.8	\$29.5	\$5.4	\$24.0	\$3.6	\$27.6	\$20.4
4	292.4	\$29.5	\$5.6	\$23.9	\$1.5	\$25.4	\$18.8
5 debt retired	290.9	\$29.5	\$5.7	\$23.8	\$1.5	\$25.3	\$0.4
6	289.5	\$29.5	\$5.9	\$23.7	\$0	\$24.5	\$0
7	288.0	\$29.5	\$6.0	\$23.5	\$0	\$23.8	\$0
8	286.6	\$29.6	\$6.2	\$23.4	\$0	\$23.0	\$0
9	285.1	\$29.6	\$6.3	\$23.3	\$0	\$22.3	\$0
10 last PTC yr	283.7	\$29.6	\$6.5	\$23.1	\$0	\$21.5	\$0
11 PTC cliff	282.3	\$19.6	\$6.6	\$12.9	\$0	\$11.3	\$0
12	280.9	\$19.6	\$6.8	\$12.8	\$0	\$11.2	\$0
15	276.7	\$19.8	\$7.3	\$12.5	\$0	\$10.9	\$0
20	269.7	\$20.2	\$8.3	\$11.9	\$0	\$10.2	\$0
25 end of life	263.1	\$20.6	\$9.4	\$11.2	\$0	\$9.5	\$0

Pre-COD PTC: \$53.74/MWh 25-year escalator: 25-yr term Base: \$45.00/MWh (2025)	IRA 45Y PTC: \$35.54/MWh 25-year escalator: 25-yr term Sponsor pays from COD Energy community bonus included	REC revenue: \$9.96/MWh 25-year escalator: 25-yr term SPP market rate
Blended revenue: \$99.24/MWh Rec'd from PTC + REC Crops to \$85.32/MWh at Yr 11	LCOE: \$49.2/MWh Precedent: NREL's ECR method \$41.60/MWh leveraged equity	With 10% equity IRR: ~35% Precedent: ECR + 10% Industry hurdle: 10%
NPV: +\$124.3M Curtailment capex + PPA + REC \$63.3M unsubsidized	WACC: 5.62% leveraged Cost of debt 7.5% leveraged 10% base reduction via TC Rpt	NPV payback: ~Year 5 Curtailment capex + PPA + REC \$63.3M unsubsidized

Revenue anchored by three streams: PPA at \$53.74/MWh (Escalated from LevelTen Nebraska/SPP P25 benchmark), IRA \$45Y PTC at \$35.54/MWh for 10 years, and RECs at \$9.96/MWh, blended Year 1 revenue of \$99.24/MWh. Project qualifies as an energy community, adding a 10% PTC bonus. Unsubsidized NPV of +\$53.3M confirms viability on fundamentals alone.

Total Pre COD CapEx: \$117.6M (\$1,626/kW)

Capital stack: 55% senior debt, 35% tax equity (IRA \$45Y PTC flip), 10% sponsor equity. Senior debt fully retired by Year 5, before the Year 11 PTC cliff, generating strong post-debt cash flows.



TrojanWind focuses heavily on a community-centric approach.

Besides a detailed community benefit plan, and powering the tax base to help stimulate the economy –a trickle that starts the steam to prevent rural emigration.

Risk Mitigation

Climatic Considerations

Antelope county experiences an estimated 15 days of icing annually. This translates to a loss of 6.54 GWh or \$440,000 yearly.

Mitigation: Implementation of the Vestas De-icing System results in estimated recovery of 70% of the energy loss; or 4.74 GWh, giving an NPV of \$1.82 million over the farms 25 year life-time.

This same energy is sufficient to power 412 homes (13.1% of Antelope county) and avoid 1600 tCO₂ annually from being released by oil or gas energy production.



Environmental Considerations

Although our selected site lies outside the Whooping Crane central flyway migration corridor and is over one mile from the Grove Lake Wildlife Management Area, a concern for the potential impact on three endangered species (Whooping Cranes, Northern Long-Eared Bat, and Tricolored Bat) remains.

Mitigation: Regular surveys conducted during construction to identify potential threats. Construction will be halted accordingly.

Implementation of the Vestas Bat Deterrent System, reducing bat mortality by up to 78%.

The RoBird™, a peregrine falcon mimicking drone, will provide a humane method to disperse birds and bats from the area at risk. Additionally, the RoBird™ can be used to conduct surveillance and inspections from above, allowing us to conduct maintenance only when strictly necessary.

AI SmartCurtailment is another innovation that helps us bring the curtailment needs down by approximately 2% (from 3.5% to 1.5%). This is achieved by only curtailing select turbines within the endangered animals' path, instead of the entire farm.

Stakeholder Engagement

As an **independent developer**, Trojan Wind identified and mapped stakeholders across four dimensions: decision authority, proximity to site, economic interest, and ecological sensitivity.

Approving Authorities

Antelope County Planning Commission (CUP), Nebraska Power Review Board (§70-1014.02 exemption), FAA (obstruction clearance), and USFWS (eagle/bat take permits) represent critical path, no project without them.

Grid & Offtake

NCCPD and SPP govern interconnection. Nebraska utility offtakers anchor the PPA at \$49.90/MWh, making early utility engagement essential to financial close.

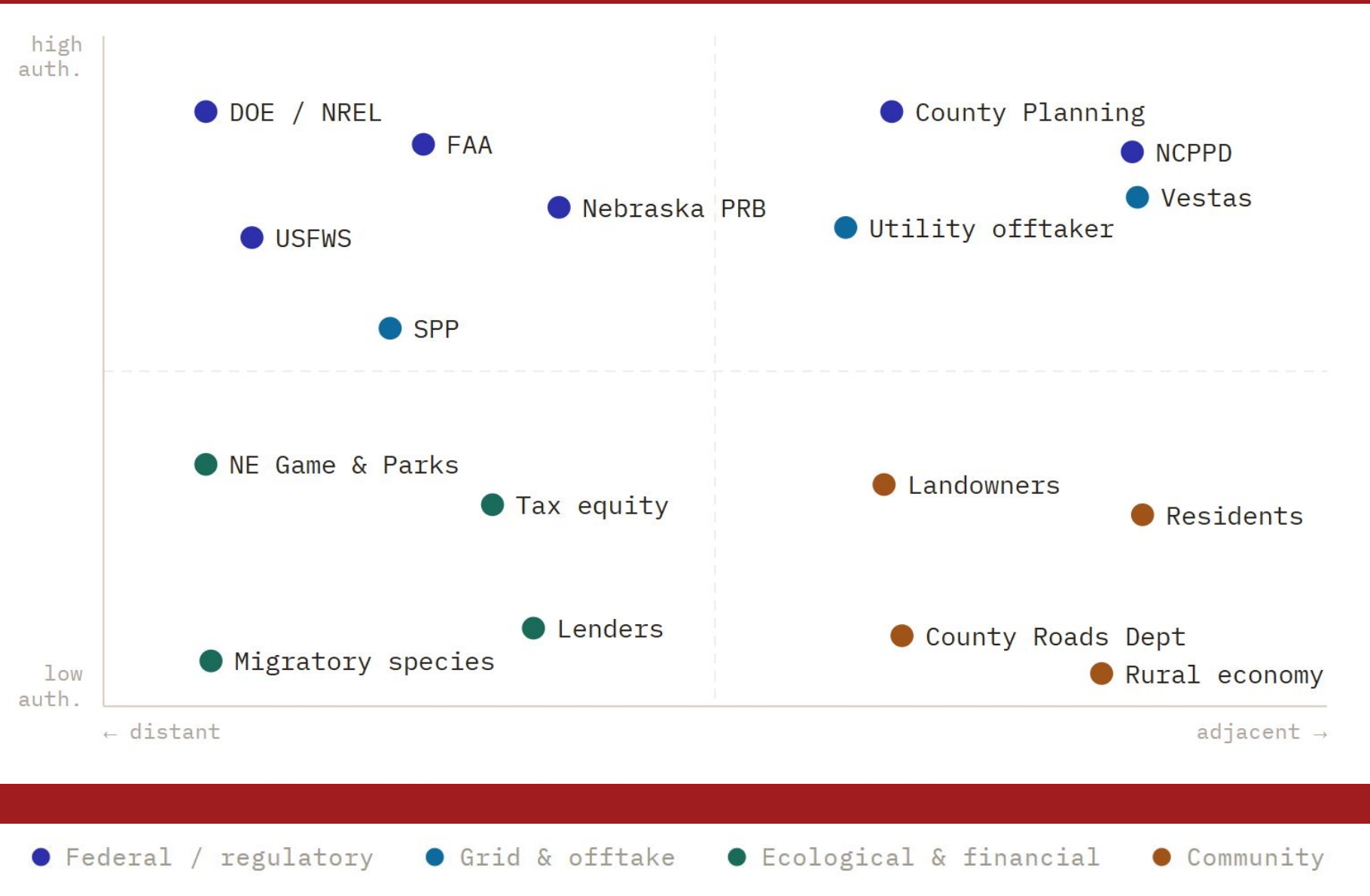
Landowners & Adjacent Community

Antelope County farmers are the project's most proximate stakeholders, lease agreements (~\$5,000–\$10,000/turbine/yr) convert potential opponents into advocates. Residents within 6 km are engaged on noise and visual impact.

Ecological Stakeholders

USFWS, Nebraska Game & Parks, and the species themselves, whooping crane, northern long-eared bat, tricolored bat, require mitigation commitments before permits are issued.

The below figure shows stakeholders organized by degree of authority and physical proximity to the proposed windfarm site.



How We Compare

	Avg US Farm	Grande Prairie	Haystack Wind	TrojanWinds
Turbine	—	V110-2.0 MW	SG 5.0 MW	V162-7.2 MW
Capacity factor	~36%	~42.8%	Not published	~49% (modelled)
Wake loss	8–10%	Not published	Not published	7.34% (modelled)
Siting method	Standard GIS	Standard GIS	Standard GIS	MCDA + ML
PTC bonus	Base only	Base only	Base only	+10% Energy Community
Debt structure	Standard	Not disclosed	Not disclosed	Retired before Yr 11 cliff
Innovation stack	None	None	30-yr Siemens O&M	AI SmartCurtailment, De-icing, drone + digital twin O&M

Trojan Wind outperforms industry averages across every key metric: a 51.6% net capacity factor and ~325 GWh/yr AEP exceed the typical Great Plains benchmark, while a leveraged LCOE of \$43.88/MWh sits below the NREL ATB \$35–\$5/MWh range midpoint, and a project IRR of ~17% with sponsor equity IRR of ~35% comfortably clears the 12% industry hurdle rate.



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